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Inventor(s)

Skinner; Gary F. et al.

System for Automatically Charging Spaced-Apart Devices

Abstract

A system for automatically charging energy storage devices associated with spaced-apart apparatuses includes a system power supply connected to a power hub, a power bus for providing power from the power hub to the energy storage devices, and charging taps for drawing power from the bus and charging the devices. Charging taps are continuously powered and may connect to a single device or more than one device. Generally only one energy storage device is charged at a time.

Inventors: Skinner; Gary F. (Broomfield, CO), MacKenzie; John J. (Wheat Ridge, CO)

Applicant: Cronopower LLC (Broomfield, CO)

Family ID: 96880518

Assignee: Cronopower LLC (Broomfield, CO)

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to apparatus and methods for automatically charging batteries or supercapacitors in spaced-apart devices. These devices typically have built-in energy storage so they can operate during a power outage. In particular, the present invention relates to charging energy storage items in motors for motorized window treatments. Other applications include emergency lighting and smoke detectors.

Discussion of Related Art

[0002] Motorized window treatments are often operated by battery powered motors. The window treatments tend to be spread out over a building and often have different motors with different charging requirements, as the windows may have different sizes and coverings. Currently, the motors are manually charged with plug in power supplies and require human intervention. Currently, motors are not using supercapacitors as they must be charged too often. Smoke detectors typically use primary cells that must be replaced yearly.

SUMMARY OF THE INVENTION

[0003] A system for automatically charging energy storage devices located at or within spaced-apart apparatuses includes a system power supply connected to a power hub, a power bus for providing power from the power hub to the energy storage devices, and charging taps for drawing power from the bus and charging the devices. Charging taps are continuously powered and may connect to a single device or more than one device.

[0004] A connecting cable may be connected between the charging tap and the device, and include a resistive jumper that tells the tap how much voltage and current the device needs to charge. This allows the system to be used with various devices that require different charging voltages or currents.

[0005] The tap has a timer for each device attached to the tap. When the timer expires, the tap determines whether the bus is free to charge the apparatus's energy storage device or if the bus is already in use. If the bus is available, the tap marks the bus as busy and converts the bus power to what the apparatus's energy storage device needs. The energy storage device is charged for a fixed amount of time, or alternatively until the associated device stops drawing current. Then the tap removes power from the energy storage device and resets the timer to schedule the next charge. The bus is now free to charge another device.

[0006] Alternately, the tap could communicate with the device, or be built into the device and then have knowledge of the battery state of charge, and only charge when needed.

[0007] Alternately, the tap could have a built-in energy storage device like a supercapacitor which is then used to run a DC powered device that does not have an internal battery. The tap can then monitor the state of charge of the storage device and charge as needed. Again, this could be built in to the device.

[0008] A system for automatically charging spaced-apart energy storage devices includes a power

hub connected to a system power supply, a plurality of energy storage devices to be charged, the energy storage devices spaced apart from each other, a bus for distributing power from the power hub, charging taps located at the energy storage devices and connected to the bus and to the energy storage devices, the charging taps configured to draw power from the bus and to charge the energy storage devices.

[0009] Each charging tap includes a processor including a timer configured to determine when an associated energy storage device is to be charged and the system is configured such that only a subset of the energy storage devices is charged at the same time. Usually only one energy storage device is charged at a time. Identifying mechanisms may be connected to each built-in energy storage device to determine the power requirements of that built in energy storage device. Then the charging taps provide power configured according to associated identifying mechanisms.

[0010] Some of the energy storage devices and their charging taps are built into their apparatuses. Some energy storage devices may located within charging taps and power their apparatuses from there. Some energy storage devices may located within apparatuses, discrete from charging taps.

[0011] The processors may further include a bus-check mechanism configured to check whether the bus is in use when a timer indicates an energy storage device is to be charged, and to delay charging that energy storage device if the bus is in use. Each timer may have a programmed time slot charging occurs during that time slot.

[0012] Usually each charging tap is connected to only one or two apparatuses.

[0013] The energy storage devices may comprise batteries or capacitors. The energy storage devices are not necessarily all similarly configured. In some cases the apparatuses include motors, for example to operate window treatments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 is a schematic block diagram showing a system for automatically charging devices with and without built-in energy storage.

[0015] FIG. 2 is a schematic block diagram of a charging tap used for charging devices with internal energy storage (e.g., batteries or supercapacitors).

[0016] FIG. 3 is a schematic block diagram of a charging tap built into a device with internal energy storage.

[0017] FIG. 4 is a schematic block diagram of a charging tap used for devices that do not have internal energy storage.

TABLE-US-00001 TABLE 1 DETAILED DESCRIPTION OF THE INVENTION

Reference number	Element
100	System for charging devices
101	System power supply
102	Power hub
105	Power bus
107	Splitter
140	Charging tap built into a device with energy storage (FIG. 3)
144	Device with a built-in tap
140	150 Charging tap for a device that has energy storage (FIG. 2)
152	Connecting cable tap 150 to device 154
154	Device with internal energy storage
160	Charging tap with energy storage for a plain device (FIG. 4)
162	Connecting cable tap 160 to device 164
164	Plain device (no internal energy storage)
210	DC to DC converter and current measurement
212	Relay
216	Transceiver
220	Processor
224	Resistor - part of cable 152
226	Connector - connecting cable 152 to charging tap 150
228	Connector - connecting cable 152 to device 154
240	Bus negative
242	Bus positive
244	Bus signal
246	User interface
248	LED
250	Button
316	Transceiver
320	Processor
329	Charge controller for storage
330	Built-in Energy Storage for device components
332	Device components
340	Bus negative
342	Bus positive
344	Bus signal
346	User interface
348	LED
350	Button
412	Relay
414	Charge controller for storage
418	416 Transceiver
418	Energy storage device built into tap
420	Processor
422	DC to DC converter and current measurement
424	Resistor - part of cable 162
427	Connector - connecting cable 162 to tap

160 Connector - connecting cable 162 to device 164 430 Negative power lead 432 Positive power lead 440 Bus negative 442 Bus positive 444 Bus signal 446 User interface 448 LED 450 Button

[0018] Table 1 shows reference numbers and their associated elements.

[0019] FIG. 1 is a schematic diagram of an example system **100** for automatically charging devices which are spaced apart. FIG. 1 shows a system for automatically combinations of charging devices **144**, **154** and **164** via charging taps **140**, **150**, and **160**. Power for the system is provided by power supply **101** drawing from mains power, which connects to power hub **102** which is connected to power bus **105** and provides power to bus **105**. Charging taps **140**, **150** and **160** connect to power bus **105** and generally are continuously powered.

[0020] System **100** may include one or more external charging taps **150** for devices **154** having their own energy storage (see FIG. 2). System **100** may include one or more charging taps **140** which are built into devices **144** which have internal energy storage (see FIG. 3). System **100** may include one or more charging taps **160** which store energy to power devices **164** that do not have their own internal energy storage (see FIG. 4). Or, system **100** may include various other configurations of charging taps. As shown in FIG. 1, different combinations of charging taps **140**, **150**, **160** can be used by the same system **100**, making the system adaptable.

[0021] Devices **154** connect to charging taps **150** via connecting cables **152**. Each device **154** does have its own energy storage device. Connecting cable **152** is configured to identify the device **154** it is attached to, and delivers the power required by device **154** for charging. See FIG. 2 for a detailed example of charging tap **150**.

[0022] Charging tap **140** is built into device **144** which has internal energy storage. See FIG. 3 for a detailed example of tap **140**.

[0023] Device **164** connects to charging tap **160** via connecting cable **162**. Device **164** in this example has no internal battery or other energy storage device, so charging tap **160** provides the energy storage capacity. Connecting cable **162** is configured to identify the device **164** it is attached to, and provides the power required by device **164**. An example of this configuration is shown in FIG. 4.

[0024] Power bus **105** in this example is an at-least-3 conductor cable connected in a daisy-chain, star or combination of these topologies. Splitters **107** may be used to help create star topologies as shown in FIG. 1. If a cable with more conductors is used, the spare conductors will generally be paralleled to reduce resistance and allow longer wires to be used. More conductors may accommodate commonly available wire such as cat5. FIG. 2 is a schematic block diagram of a charging tap **150**, attached to a device **154**, via connecting cable **152**. Device **154** has internal energy storage (not shown). Charging tap **150** includes DC-DC converter with current measuring **210**, relay **212**, transceiver **216**, and processor **220**. Power bus **105** provides power and carries a bus signal. In this example, **240** is bus negative, **242** is bus positive, and **244** is signal (optional).

[0025] Charging tap **150** uses DC-DC converter **210** to convert voltage from power bus **105** to the voltage needed by device **154** to charge its internal energy storage.

[0026] Each connecting cable **152** is configured to identify its particular attached device **154**, so the required power is provided to that device **154**. In this embodiment, connecting cable **152** includes a resistor **224** which signals the appropriate power to be provided to each device **154** at appropriate intervals to keep the internal storage charged. Device **154** can be various brands with different charging and voltage requirements. Charging tap **150** may be attached to more than one device **154**, but generally will only charge one device **154** at a time. This reduces the size of power supply **101** required.

[0027] In this example, cable **152** is connected during installation and is not intended to be regularly unplugged. Cable **152** provides two conductors for charging the energy storage in device **154**. Connector **228** is configured and selected depending on what brand of device **154** it is being connected to. Connector **226** has multiple pins with a method for identifying which cable **152** is

being used and thus determining, for example, what brand and model of device **154** is attached to determine the appropriate voltage and current necessary for charging. There are many ways to achieve this, but this example implementation uses resistor **224** between 2 pins independent of the 2 wires providing power. Processor **220** measures the resistance and sets the output voltage of DC-DC converter **210** before activating optional relay **212** (or equivalent depending on which device is being charged). Relay **212** may be used if it is useful to electrically isolate device **154** from the power bus when it is not being charged.

[0028] The dashed line in cable **152** represents possible conductors for feedback from device **154** indicating the state of its internal charge, and possibly other data or control signals. This signaling could be implemented via multiple protocols, with USB being convenient as the connector to many devices is already USB. This signal can be used to stop charging when device **154** is complete. This feedback could also be done via a wireless protocol between the device **154** and the tap **150**, or via a bridge device in the hub **102**.

[0029] FIG. **3** is a schematic block diagram of charging tap **140** built into a device **144** which has internal energy storage **330**. The tap **140** includes charge controller **329**, transceiver **316**, and processor **320**. Power bus **105** provides power and carries a bus signal. In this example, **340** is bus negative, **342** is bus positive, and **344** is signal.

[0030] Processor **320** uses charge controller **329** to convert voltage from power bus **105** to the voltage needed by energy storage **330**. Energy storage **330** provides power to the device components **332** that implement the functions of device **144**, e.g., motor, light or sensor.

[0031] FIG. **4** is a schematic block diagram of charging tap **160** with energy storage **418** for a plain attached device **164**, which does not have internal energy storage, via connecting cable **162**. Tap **160** includes charge controller **414**, DC-DC converter **422**, energy storage **418**, relay **412**, transceiver **416**, and processor **420**. Power bus **105** provides power and carries a bus signal. In this example, **440** is bus negative, **442** is bus positive, and **444** is signal (optional).

[0032] Tap **160** uses charge controller **414** to convert voltage from power bus **105** to the voltage needed to charge its internal energy storage **418**. Tap **160** uses DC-DC converter **422** to convert voltage from energy storage **418** to power device **164** which generally does not have internal energy storage.

[0033] Each connecting cable **162** is configured to identify its particular attached device **164**, so the required power is provided to that device. In this embodiment, connecting cable **162** includes a resistor **424** which signals the appropriate voltage to be provided to each device **164**. Device **164** can be various devices or brands with different voltage requirements. Tap **160** may be attached to more than one device **164**.

[0034] In this example, cable **162** is connected during installation and is not generally unplugged. Cable **162** provides two conductors **430**, **432** for supplying power to device **164**. Connector **428** is configured and selected depending on what brand of device **164** it is being connected to. Connector **427** has multiple pins with a method for identifying which cable **162** is being used and thus determining what brand and model of device **164** is attached to determine the appropriate voltage necessary for operation. There are many ways to achieve this, but this example implementation uses resistor **424** between 2 pins independent of the 2 power wires **430**, **432**. Processor **420** measures the resistance and sets the output voltage of DC-DC converter **422**.

[0035] Power supply **101** provides sufficient voltage and current to be able to charge any supported brand of device **154**, or devices **144** or **164**. This actual value depends on how DC-DC converter **210** or **422** or charge controller **329** or **414** is implemented. Generally, only one device is charged at a time, so power supply **101** only needs to be large enough for that demand, plus the small idle current to keep each charging taps **140**, **150** or **160** running. Some applications may benefit from powering a few devices at the same time (e.g. two). One particular implementation uses a buck converter, so power supply **101** is 24V 1A which accommodates 16.8V devices at approximately 1A. A buck/boost converter could be used to accommodate higher voltage devices **154**, **164**.

[0036] Charging only one device at a time is accomplished in this example using a busy signal on power bus **105** and taps **140, 150, 160** waiting when bus **105** is busy. An alternative method would be to give each tap **140, 150, 160** a time slot to do its charging. This utilizes a method to synchronize all taps so each one knows when it is their turn.

[0037] The batteries in window covering motors are typically Li-ion, but this system can be used to charge any battery type assuming that device **154** (for example) has an appropriate charge controller built in. This system **100** can support solid state batteries or other chemistries as they become available. This system can also allow devices to use supercapacitors for charge storage by decreasing the time between charges appropriately.

[0038] Another useful implementation would be to have most of power bus **105** on one type of cable and interconnect with a connection adapter (not shown) to convert to a smaller connector (not shown) used on window covering motors where the motor connector is dual purpose. E.g., a USB C connector on motor **144** could allow the standard USB power pins to be used to charge the motor if it is not connected to power bus **105**. Alternate pins on the USB connector could be used when connected to power bus **105**. This allows one device to be sold for either use case.

[0039] At least two of the bus **105** conductors provide power (**240, 242, 340, 342, 440, 442**) to taps **140, 150** and **160** and one or more conductors (**244, 344, 444**) may be used for signaling between the taps and power hub **102**. Signaling on power bus **105** may use any existing protocol and transceiver **216, 316, 416**, e.g., LIN or RS485, or may use a custom system. This implementation uses a LIN transceiver as it only needs 1 wire and accommodates 24V signaling.

[0040] Alternately transceiver **216, 316, 416** could be wireless. This would reduce the required number of wires to two.

[0041] In some embodiments, processor **220, 320, 420** has a timer (not shown, preferably kept in non-volatile memory to survive power outages) for each connected device to be charged **144, 154, 160** and internal energy storage **330, 418**. When that timer expires (for example every 3 months), Processor **220, 320, 420** determines if the bus **105** is free to use to charge. Determination of busy can be accomplished with messages using a multidrop protocol, or some custom protocol. [0042] If bus **105** is in use, then processor **220, 320, 420** will wait until the bus is idle. [0043] If bus **105** is idle, then processor **220, 320, 420** will mark bus **105** as busy, and then apply power to the device **154** or internal energy storage **330, 418** that timed out as described earlier. Generally, it also starts a maximum charge timer.

[0044] As an example, refer to FIG. 2. Processor **220** in tap **150** checks the amount of current being sourced by DC-DC converter **210**, and will remove power if device **154** stops drawing current because it is sufficiently charged, or if the maximum charge timer runs out. Then processor **220** assumes that device **154** is charged, and resets the timer for the next charge (typically 3 months). Taps **140** or **160** have built-in energy storage **330, 418**, so can easily determine state of charge and handle recharging based on that rather than the simple timer.

[0045] The optional user interface **246, 346, 446** including for example button **250, 350, 450** and LED **248, 348, 448** shown may be used to administer intervals, to check times till next charge, to cause immediate charge cycle when needed, and to aid in installation.

[0046] Multiple taps with internal energy storage **160** or without **150** or built in **140** may all be on power bus **105** simultaneously and the system will generally only charge one device **154** or one internal energy storage element **418** or **330** at a time.

[0047] Many variations and alternate implementations fall within the spirit of this invention.

[0048] Supercapacitors are generally not currently used to power motors in stand-alone window motor applications because they self-discharge and the user would be charging them way too often to be practical. Charging tap **150** at short intervals overcomes this problem and can allow a motor manufacturer to use supercapacitors inside their motor.

[0049] The bus signal **244, 344, 444** on Power bus **105** could be extended to include messages for controlling the device or data collection.

[0050] While the exemplary preferred embodiments of the present invention are described herein with particularity, those skilled in the art will appreciate various changes, additions, and applications other than those specifically mentioned, which are within the spirit of this invention. For example, if the busy signal is sent wirelessly, the power hub may be a simple connector block.

Claims

1. A system for automatically charging spaced-apart energy storage devices comprising: a power hub connected to a system power supply; a plurality of energy storage devices to be charged, the energy storage devices spaced apart from each other; a bus for distributing power from the power hub; charging taps located at the energy storage devices and connected to the bus and to the energy storage devices, the charging taps configured to draw power from the bus and to charge the energy storage devices; wherein each charging tap includes a processor including a timer configured to determine when an associated energy storage device is to be charged; and wherein the system is configured such that only a subset of the energy storage devices is charged at the same time.
 2. The system of claim 1 wherein only one energy storage device is charged at a time.
 3. The system of claim 1 wherein some energy storage devices and charging taps are built into spaced-apart apparatuses.
 4. The system of claim 3 further comprising an identifying mechanism connected to each built-in energy storage device to determine the power requirements of that built in energy storage device.
 5. The system of claim 1 wherein some energy storage devices are located within charging taps and wherein apparatuses are powered by those charging taps.
 6. The system of claim 5 further comprising an identifying mechanism connected to each energy storage device configured to determine the power requirements of that energy storage device.
 7. The system of claim 1 wherein some energy storage devices are located within apparatuses discrete from charging taps.
 8. The system of claim 7 further comprising an identifying mechanism connected to each apparatus configured to determine the power requirements of the energy storage device within the connected apparatus.
 9. The system of claim 8 wherein the charging taps provide power configured according to associated identifying mechanisms.
 10. The system of claim 7 wherein the processors further include a bus-check mechanism configured to check whether the bus is in use when a timer indicates an energy storage device is to be charged, and to delay charging that energy storage device if the bus is in use.
 11. The system of claim 10 wherein each timer has a programmed time slot and wherein an associated energy storage device is charged during that time slot.
 12. The system of claim 7 wherein each charging tap is connected to only one or two apparatuses.
 13. The system of claim 1 wherein the energy storage devices comprise batteries.
 14. The system of claim 1 wherein the energy storage devices comprise capacitors.
 15. The system of claim 1 wherein the apparatuses include motors.
 16. The system of claim 15 wherein the motors operate window treatments.
 17. The system of claim 1 where the energy storage devices are not all similarly configured.
 18. The system of claim 1 wherein the processors further include a bus-check mechanism configured to check whether the bus is in use when a timer indicates an energy storage device is to be charged, and to delay charging that energy storage device if the bus is in use.
 19. The system of claim 18 wherein each timer has a programmed time slot and wherein an associated energy storage device is charged during that time slot.
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